Unit 11, Lesson 4: Solving Problems Involving Quadrilaterals Inscribed in Circles

Benchmark

MA.912.GR.6.3 Solve mathematical problems involving triangles and quadrilaterals inscribed in a circle.

Learning Target

- ☐ I can solve mathematical problems involving quadrilaterals inscribed in a circle.

11.4.1 Warm-Up: Notice and Wonder

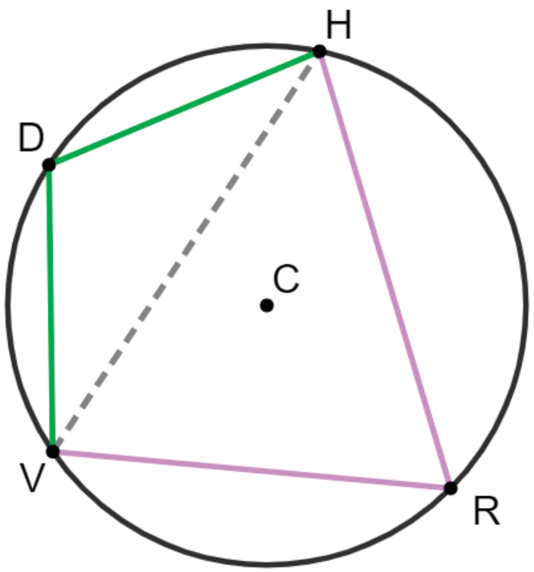
Machu Picchu is an Inca citadel in southern Peru. Machu Picchu was built in the classical Inca style with polished dry-stone walls and consists of more than 150 buildings. Although many of the stones that were used to build the city were more than 50 pounds, it is believed that no wheels were used to transport these rocks up the mountain.

1. What do you notice?

2. What do you wonder?

11.4.2 Exploration: Quadrilaterals Inscribed in Circles

1. Circle C is shown, where $\overline{DV} \cong \overline{DH}$. $m\,\widehat{RV} = 100°$, $m\,\widehat{DV} = 68°$, $m\,\widehat{HDV} = 136°$, and $m\,\widehat{HRV} = 224°$.



a. Complete the table.

Angle Measures	Angle Measures
$m\angle RHV =$	$m\angle HDV =$
$m\angle DHV =$	$m\angle DVH =$
$m\angle HRV =$	$m\angle HVR =$

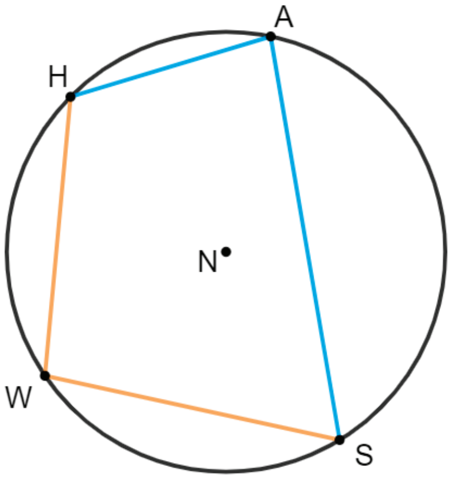
b. Work with your partner to make a conjecture about the relationships of the angles in $DHRV$.

c. Ask two classmates for their conjectures. Record their conjectures and write your summary of their reason. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

Conjecture	My Summary of Their Reason	Initials

d. Review your conjecture with your partner. If necessary, make any changes.

2. Circle N has points H, A, S, and W on the circle. Quadrilateral $HASW$ is inscribed in the circle. $m\,\widehat{HA} = 59°$, $m\,\widehat{HWS} = 166°$ and $m\,\widehat{WHA} = 136°$.



Use your conjecture to find the measures.

Angle Measures	**Arc Measures**
$m\angle WHA =$	$m\,\widehat{SA} =$
$m\angle HAS =$	$m\,\widehat{SW} =$
$m\angle ASW =$	$m\,\widehat{HW} =$
$m\angle SWH =$	

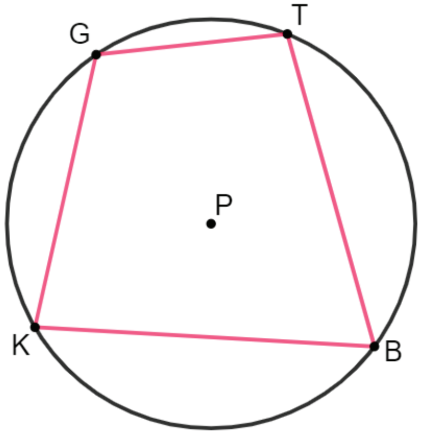
11.4.3 Guided Practice: Quadrilaterals Inscribed in Circles

1. Complete the statement.

If a quadrilateral is inscribed in a circle,
- ○ adjacent
- ○ opposite

angles are
- ○ congruent.
- ○ complementary.
- ○ supplementary.

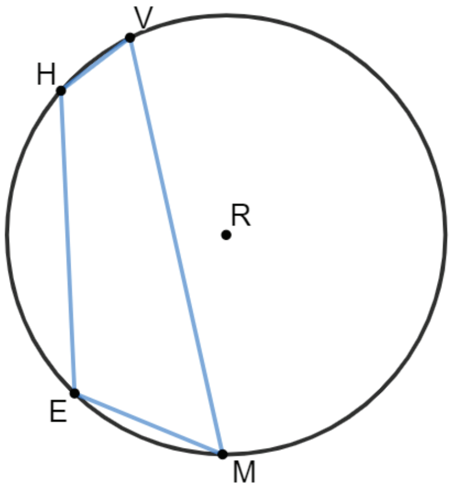
2. Circle P has $GTBK$ inscribed in the circle. $m\,\widehat{TBK} = 206°$ and $m\angle GKB = 84°$.



a. What is the $m\angle TBK$?

b. What is the $m\angle GTB$?

3. Circle R has $VMEH$ inscribed in the circle. $m\angle HEM = (2x - 7)°$, $m\angle EHV = (2x + 3)°$, and $m\angle HVM = (x + 4)°$.



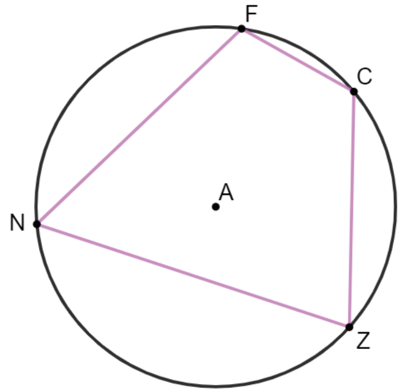
a. Find the value of x.

b. Complete the table.

$m\angle HVM =$
$m\angle HEM =$
$m\angle EHV =$
$m\angle EMV =$

11.4.4 Your Turn!

1. Circle A has $NFCZ$ inscribed in the circle. $m\angle NFC = (6x + 32)°$, $m\angle FCZ = (6y - 8)°$, $m\angle NZC = 73°$, and $m\angle FNZ = (3y - 1)°$.



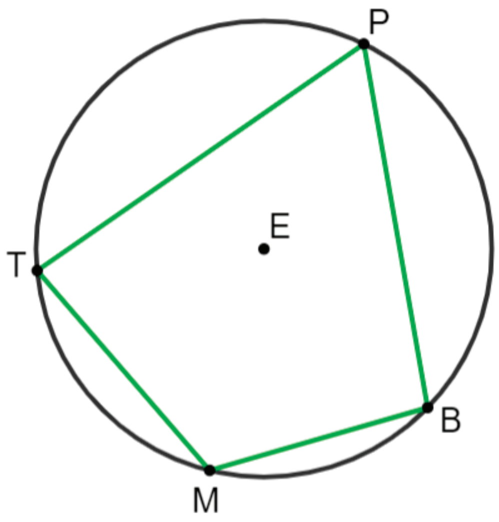
a. Find the value of x.

b. What is the $m\angle NFC$?

c. Find the value of y.

d. What is the $m\angle FCZ$?

2. Circle E has $TPBM$ inscribed in the circle. $m\widehat{TM} = 71°$, $m\widehat{MB} = 61°$, and $m\angle TMB = (12x - 3)°$.



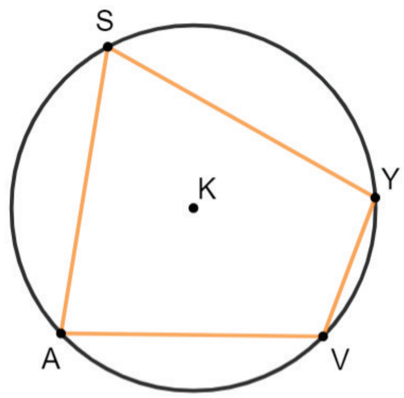
a. Find the value of x.

b. What is the $m\angle TMB$?

11.4.5 Wrap-Up: Error Analysis

1. $SYVA$ is inscribed in Circle K, where $m\angle ASY = (3x + 23)°$ and $m\angle AVY = (6x + 13)°$.

   Anastasia is finding the value of x. Her work is shown.



Anastasia's Work
$3x + 23 = 6x + 13$ $3x = 10$ $x = \frac{10}{3}$

a. Write a note to Anastasia explaining her mistake.

b. Find the value of x.